Accelerating Quantum LDPC Code Optimization Through Adversarial Games

Dayeon Yoo and Milad Marvian
Department of Electrical and Computer Engineering, University of New Mexico
Center for Quantum Information and Control (CQuIC)

qLDPC code optimization

Why optimize finite-size codes?

qLDPC codes can achieve high encoding rates with sparse, low-weight checks. However, different finite-size codes within the same family can have different performance.

Goal: fast optimization techniques for $N \sim$ tens of thousands qubits

What makes a good finite-length qLDPC code?

$\uparrow k/n$ $\quad\uparrow d$ $\quad\downarrow$ logical error rate $\quad\downarrow$ check weights

Why is qLDPC code optimization hard?

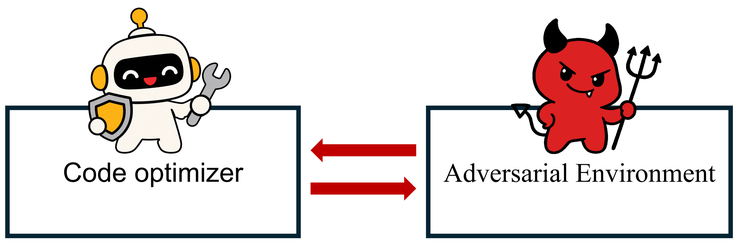
Huge search space + global metrics provide little guidance

Distance alone provides limited guidance, while LER is expensive to estimate; neither identifies which code modification is promising

Adversarial Search for Quantum Codes

Idea

We iteratively identify the current minimum-weight undetectable errors and modify the code to make as many of them detectable as possible.



- Generate modifications that make many errors detectable
- Update the code to make many identified errors detectable
- Identify minimum-weight undetectable errors

References

[1] J.-P. Tillich and G. Zémor, Quantum ldpc codes with positive rate and minimum distance proportional to the square root of the blocklength (2014)

[2] B. C. A. Freire, N. Delfosse, and A. Leverrier, Optimizing hypergraph product codes with random walks, simulated annealing and reinforcement learning (2025)

Acknowledgments

D.Y. is supported by a Translational Fellowship from QNM-I. Additional support is provided by NSF CAREER Award No. CCF-2237356.

Applying to Hypergraph Product Codes

Hypergraph-product (HGP) codes construct a CSS quantum code from two classical binary codes [1]. A symmetric HGP code is constructed by one classical parity-check matrix $H \in \mathbb{F}_2^{m\times n}$:

$$H_X = \left[H \otimes I_n \;\middle|\; I_m \otimes H^T\right], \quad H_Z = \left[I_n \otimes H \;\middle|\; H^T \otimes I_m\right]$$

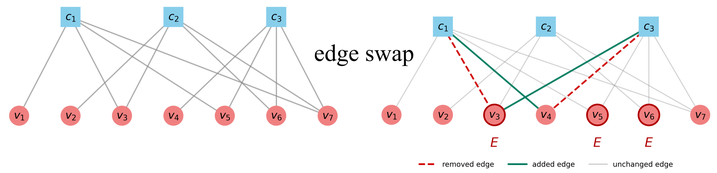
which satisfy the commutation condition $H_X \cdot H_Z^T = 0$.

For a full-row-rank parent H, the symmetric HGP code parameters are

$$[n, k, d] \longrightarrow [[n^2 + m^2, k^2, d]]$$

Thus, we can optimize the much smaller classical parent code rather than the full quantum code.

Choice of actions on Tanner graphs



Tanner graph representation of [7,4,3] classical parent code, a minimum-weight codeword, and the action that makes the undetectable error detectable.

For HGP codes, the undetectable error set consists of the weight-d codewords in the kernel of the classical parent parity-check matrix H:

$$T_d(H) = \{x \in \ker H \setminus \{0\} : \mathrm{wt}(x) = d\}$$

Suppose we have five weight-d codewords, then the undetectable-error set is

$$T_d(H) = \{t_1, t_2, t_3, t_4, t_5\}$$

	t_1	t_2	t_3	t_4	t_5	Errors made detectable
Action A	✓	✓		✓		3/5
Action B		✓	✓			2/5
Action C	✓	✓	✓	✓	✓	**5/5**

An undetectable error becomes detectable when its syndrome changes from zero to nonzero. If all weight-d codewords are removed without creating new codewords of weight $\leq d$, the distance increases.

Numerical Results

Initial code	Optimized code	$A^{\text{init}}_{d_{\text{init}}}$	$A^{\text{opt}}_{d_{\text{opt}}}$	Runtime
$[[625, 25, \mathbf{6}]]$	$[[625, 25, \mathbf{9}]]$	1	8	3s
$[[900, 36, \mathbf{10}]]$	$[[900, 36, \mathbf{10}]]$	20	10	9s
$[[1225, 49, \mathbf{6}]]$	$[[1225, 49, \mathbf{11}]]$	2	12	33s
$[[1600, 64, \mathbf{6}]]$	$[[1600, 64, \mathbf{12}]]$	1	22	34s
$[[2025, 81, \mathbf{10}]]$	$[[2025, 81, \mathbf{12}]]$	8	3	1m 18s
$[[2500, 100, \mathbf{10}]]$	$[[2500, 100, \mathbf{14}]]$	2	48	59s
$[[1224, 144, \mathbf{6}]]$	$[[1224, 144, \mathbf{8}]]$	3	26	3m 16s
$[[1125, 225, \mathbf{4}]]$	$[[1125, 225, \mathbf{6}]]$	7	9	1h 16m 42s

Here, A_d denotes the number of codewords at the current minimum weight d.

- Most searches finished in seconds to minutes.

Representative result: $[[625, 25]]$

Adversarial search		BP+OSD random-walk baseline∗
3 s	**vs.**	**49 h**
d = 9		d = 8

∗Our LER-in-the-loop implementation adapted from [2].

No LER estimation is required in the optimization loop, avoiding a major computational bottleneck.

Initial $d = 6, A_6 = 1$ → $d = 8, A_8 = 5$ → $d = 8, A_8 = 2$ → $d = 9\ A_9 = 10$

Progress is visible even when d is unchanged: A_8: $5 \rightarrow 2$.

LER (initial → optimized): $11.986 \times 10^{-3} \rightarrow 3.749 \times 10^{-3}$
68.7% reduction

BP+OSD, independent bit-flip noise, $p = 0.03$

Outlook

Beyond HGP. The same adversarial search can also be applied to BB (Bivariate Bicycle) codes.

	BB codes (Bivariate Bicycle)
undetectable errors	Sampled minimum-weight logical operators
actions	Polynomial modifications

- Evaluate the optimized codes under circuit-level noise.
- Extend the framework beyond quantum memory to fault-tolerant logical computation.